# Akoizo Original Practice - 2027 Rotation Form 3 - Answer key

### Division C | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Two second-order streams create…**

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Second order

B. Fourth order

C. First order

D. Third order

**Correct: D**

**2. (2 points) Hydrograph area represents…**

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Only peak stage

B. Only velocity

C. Only basin area

D. Volume

**Correct: D**

**3. (2 points) Impervious connectivity can affect…**

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only mineral hardness

B. Only gene frequency

C. No water process

D. Runoff timing

**Correct: D**

**4. (2 points) Groundwater divides always match surface divides?**

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Always

B. Only on maps

C. Only in winter

D. No

**Correct: D**

**5. (2 points) A 100-year label means exact spacing?**

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Always

B. Only for rivers

C. Only after one flood

D. No

**Correct: D**

**6. (2 points) Load rate needs…**

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only pH

B. Only map area

C. Only color

D. Concentration and discharge

**Correct: D**

**7. (2 points) Radial drainage can suggest…**

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Only flat sediment

B. Only a straight fault

C. No topography

D. An elevated central feature

**Correct: D**

**8. (2 points) Stage and discharge are…**

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Identical units

B. Always equal numbers

C. Unrelated by any model

D. Related but different quantities

**Correct: D**

**9. (2 points) Before/after change proves causality alone?**

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Always

B. Only after one storm

C. Only in a reservoir

D. No

**Correct: D**

**10. (4 points) Why specify the outlet?**

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

answer: It defines the contributing basin and network considered. (4 points).

**11. (2 points) A freshwater catchment receives 130 mm precipitation, loses 40 mm to evapotranspiration and exports 30 mm runoff. Find storage change.**

answer: 60 mm (2 points).

Numerical target 60; absolute tolerance 0.12. Accepted units: , mm.

**12. (2 points) A rectangular stream is 7 m wide and 2 m deep with mean velocity 0.5 m/s. Calculate discharge.**

answer: 7 m^3/s (2 points).

Numerical target 7; absolute tolerance 0.014. Accepted units: , m^3/s.

**13. (2 points) A river channel is 15 km long between points 11 km apart along the valley. Find sinuosity.**

answer: 1.3636363636363635 (2 points).

Numerical target 1.3636364; absolute tolerance 0.0027272727. Accepted units: , .

**14. (2 points) A water table drops 2 m over 100 m horizontal distance. Find hydraulic gradient.**

answer: 0.02 (2 points).

Numerical target 0.02; absolute tolerance 4e-05. Accepted units: , .

**15. (2 points) Use Q=K A i with K=0.001 m/s, A=23 m^2 and i=0.02. Find saturated Darcy discharge.**

answer: 0.00046 m^3/s (2 points).

Numerical target 0.00046; absolute tolerance 9.2e-07. Accepted units: , m^3/s.

**16. (2 points) A lake stores 3000 m^3 and exports 300 m^3/day in a steady well-mixed teaching model. Estimate residence time as storage/outflow.**

answer: 10 days (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , days.

**17. (2 points) An urbanized watershed has more impervious area. Predict one hydrograph change under a comparable storm and explain.**

answer: Faster runoff routing can shorten lag time and increase peak discharge; drainage infrastructure, storage and storm properties must be controlled. (2 points).

**18. (2 points) A stream bends through its floodplain. Locate likely erosion and deposition relative to the bend.**

answer: Erosion is commonly concentrated on the outer cut bank and deposition on the inner point bar, with local flow conditions controlling details. (2 points).

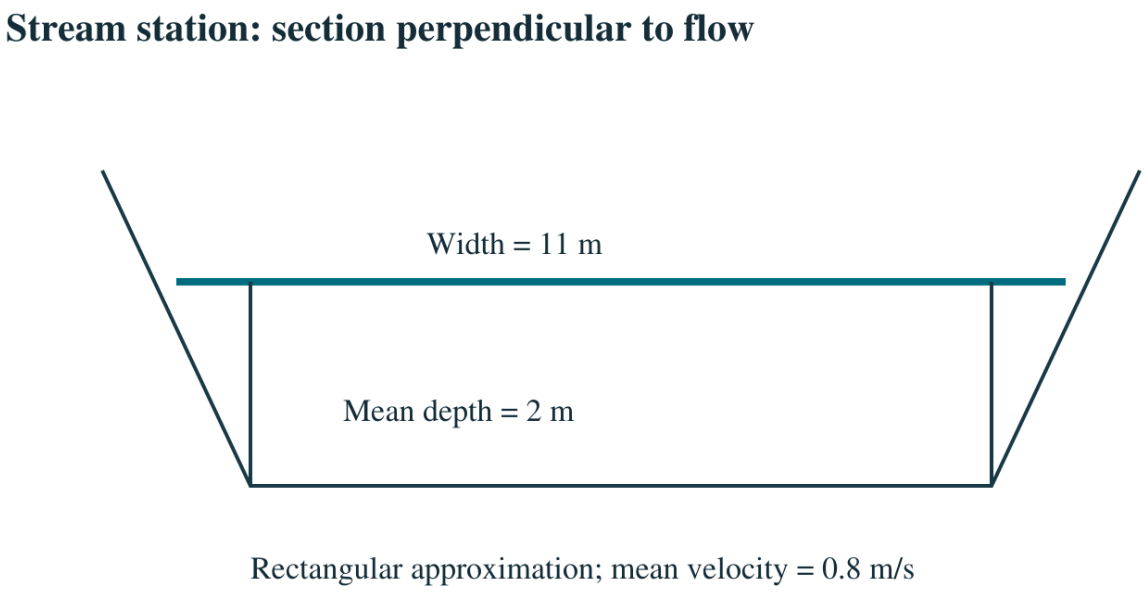
**19. (2 points) Two first-order channels join, and no higher-order channel enters. What Strahler order results, and why?**

answer: Second order; joining two channels of the same order increases order by one. (2 points).

**20. (2 points) Groundwater pumping increases near a connected river. Explain a possible effect on river baseflow.**

answer: Lower groundwater head can reduce discharge to the river or induce infiltration from it, depending on hydraulic connection and gradients. (2 points).

**21. (2 points) Use Figure F1. Estimate wetted cross-sectional area using the indicated rectangular approximation.**




F1. Stream cross section and measured velocity (original synthetic teaching figure)

answer: 22 m^2 (2 points).

Numerical target 22; absolute tolerance 0.066. Accepted units: , m^2.

**22. (2 points) Use Figure F1. Calculate stream discharge from this section.**

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

answer: 17.6 m^3/s (2 points).

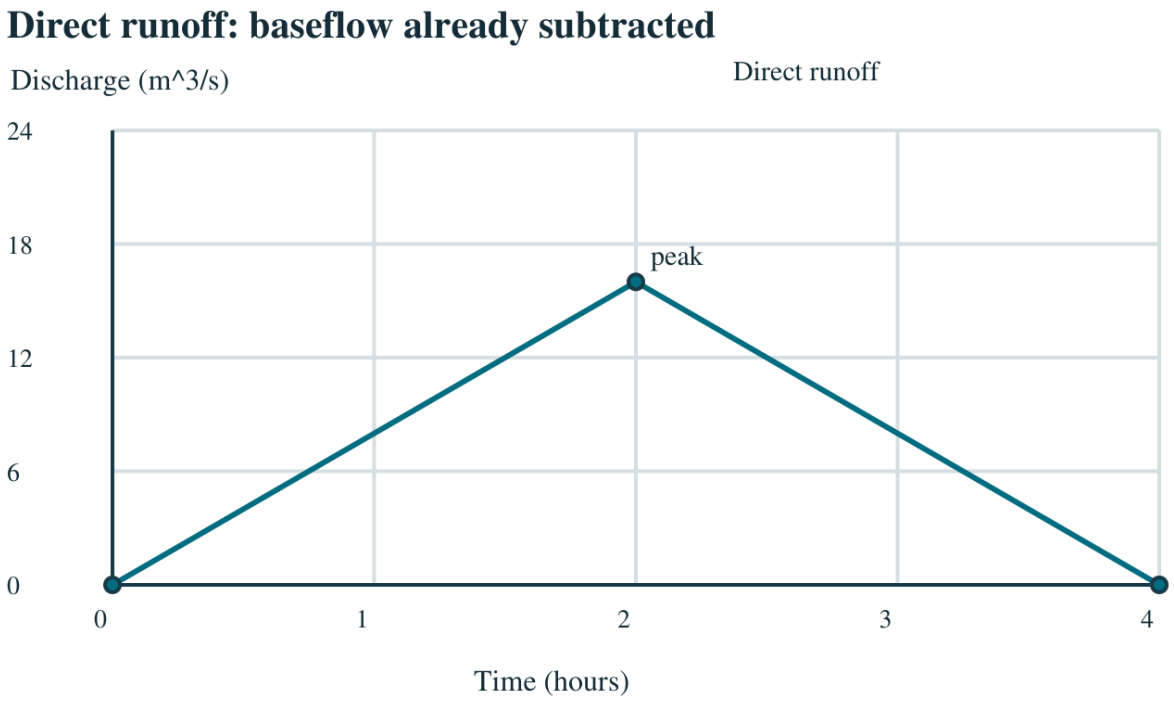
Numerical target 17.6; absolute tolerance 0.0528. Accepted units: , m^3/s.

**23. (2 points) Use Figure F1. The water rises while width and mean velocity remain fixed. Explain the predicted discharge change.**

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

answer: Greater depth increases cross-sectional area, so Q=Av increases under those assumptions. (2 points).

**24. (2 points) Use Figure F2. Read the time of peak direct runoff.**




F2. Storm hydrograph (original synthetic teaching figure)

answer: 2 hours (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: , hours.

**25. (2 points) Use Figure F2. Find the event runoff volume. Convert hours to seconds.**

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

answer: 115200 m^3 (2 points).

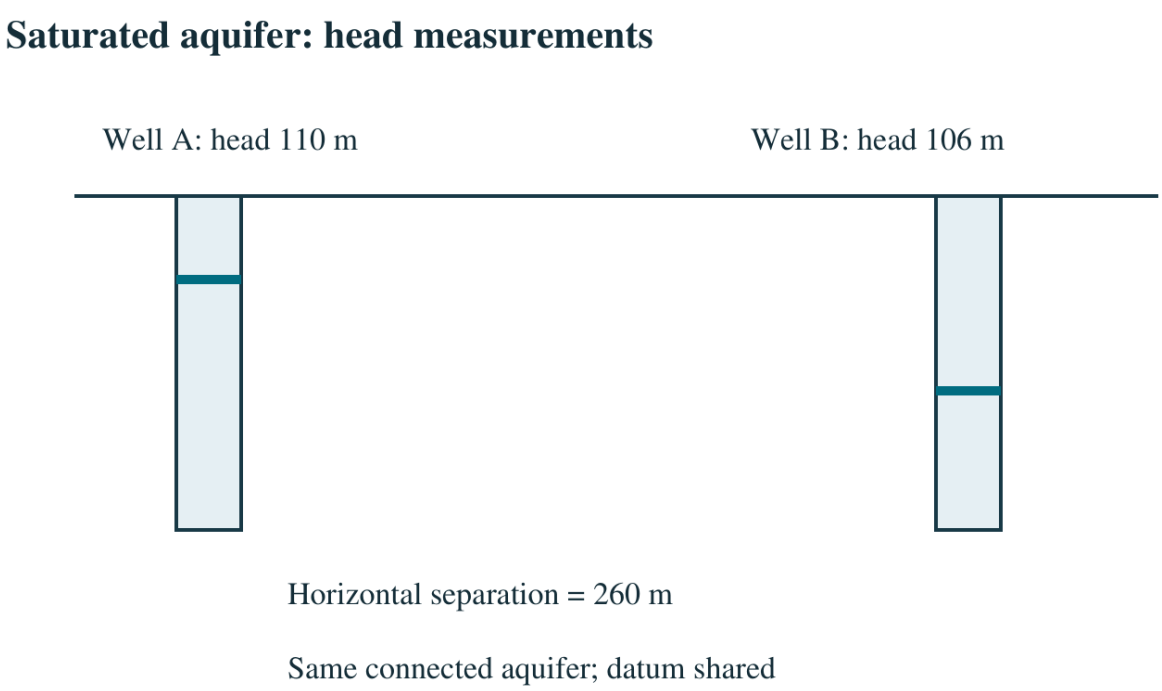
Numerical target 115200; absolute tolerance 345.6. Accepted units: , m^3.

**26. (2 points) Use Figure F2. A wetland spreads the same runoff volume over more time. Describe how its hydrograph could differ.**

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

answer: Its peak can be lower and its duration longer while the area under the discharge-time curve remains the same. (2 points).

**27. (2 points) Use Figure F3. Calculate hydraulic-gradient magnitude.**




F3. Groundwater heads in two wells (original synthetic teaching figure)

answer: 0.015384615 (2 points).

Numerical target 0.015384615; absolute tolerance 4.6153846e-05. Accepted units: .

**28. (2 points) Use Figure F3. Predict the horizontal flow direction and identify the controlling measurement.**

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

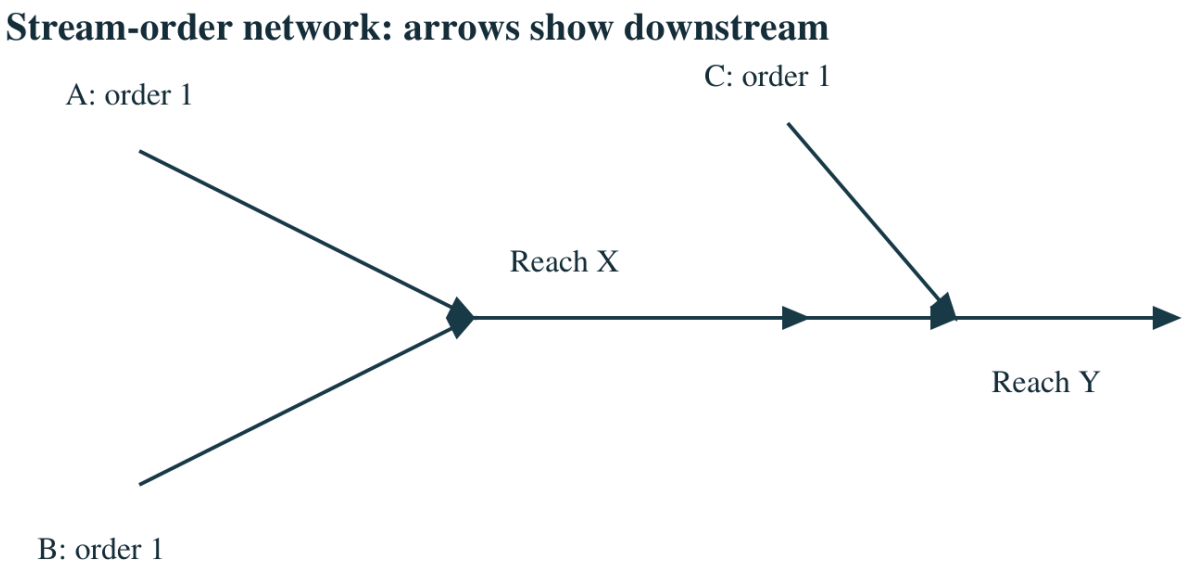
answer: From A toward B, from higher hydraulic head to lower head, assuming the connected aquifer shown. (2 points).

**29. (2 points) Use Figure F3. Why cannot the diagram alone determine groundwater discharge?**

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

answer: Hydraulic conductivity and flow cross-sectional area are also required for Darcy discharge; head gradient alone is insufficient. (2 points).

**30. (2 points) Use Figure F4. Give the Strahler order of reach X.**




F4. Tributary network (original synthetic teaching figure)

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

**31. (2 points) Use Figure F4. Give the Strahler order of reach Y.**

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

**32. (2 points) Use Figure F4. If C were second order instead, how would Y change?**

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

answer: Two second-order reaches joining would produce a third-order Y. (2 points).